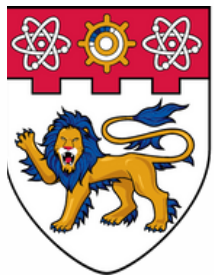




INTERPRETABLE FUZZY DEEP TRANSFORMER NEURAL NETWORK AND MULTI-AGENT REINFORCEMENT LEARNING PORTFOLIO OPTIMISATION



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Introduction

Forecasting stock prices is complex due to volatile, non-linear market dynamics. This project applies an Interpretable Deep Fuzzy Transformer Neural Network (IFDTNN) and a dynamic multi-agent reinforcement learning system for adaptive, explainable, and high-performing portfolio optimization

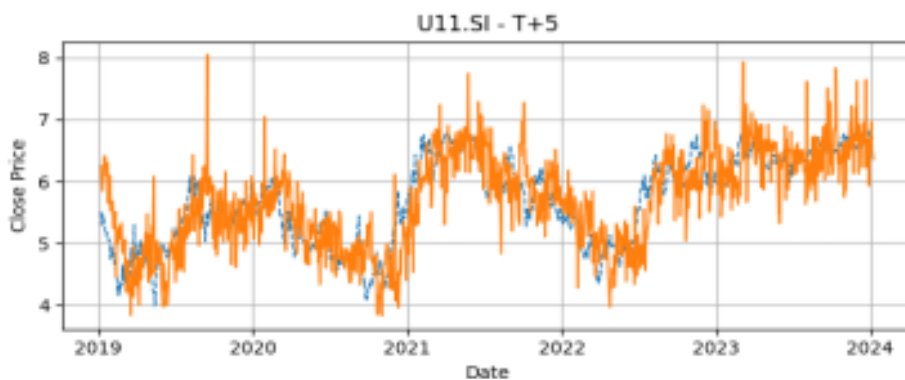
Motivation

Different stocks exhibit unique behaviors, and no single model can consistently outperform across market conditions this work leverages diverse agent architectures to maximize returns and interpretability through fuzzy logic to build stakeholder trust

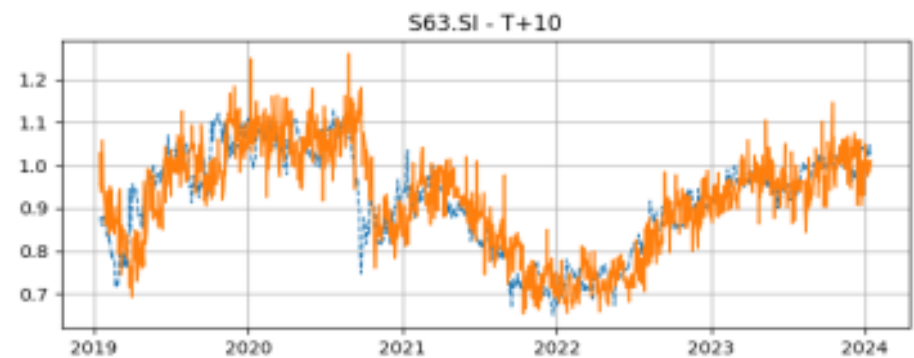
Abstract

- **Stock Price Prediction**

Custom Interpretable Fuzzy Deep Transformer model (IFDTNN) achieves 0.98+ R^2 over a 13-day forecast horizon on SGX stocks



U11.SI Actual Close Price (blue) vs Predicted Close price (T+5) (orange)



S63.SI Actual Close Price (blue) vs Predicted Close price (T+10) (orange)



Abstract

- **Interpretability**

Fuzzy rule-tagging system enhanced with Hebbian learning associates fuzzy inputs with outputs, generating clear, interpretable decision rules.

Model	Rank	Rule	
		Antecedents	Consequent
AJBU.SI	1	• $t+1_Price_c3$ • $t+4_Price_c3$ • $t+7_Vol_c0$ • $t+6_Vol_c6$ • $t+2_Price_c2$ • $t+9_ROC_c3$	$t+13_Price_c3$

AJBU.SI Top Fuzzy Rules of t+13 Models



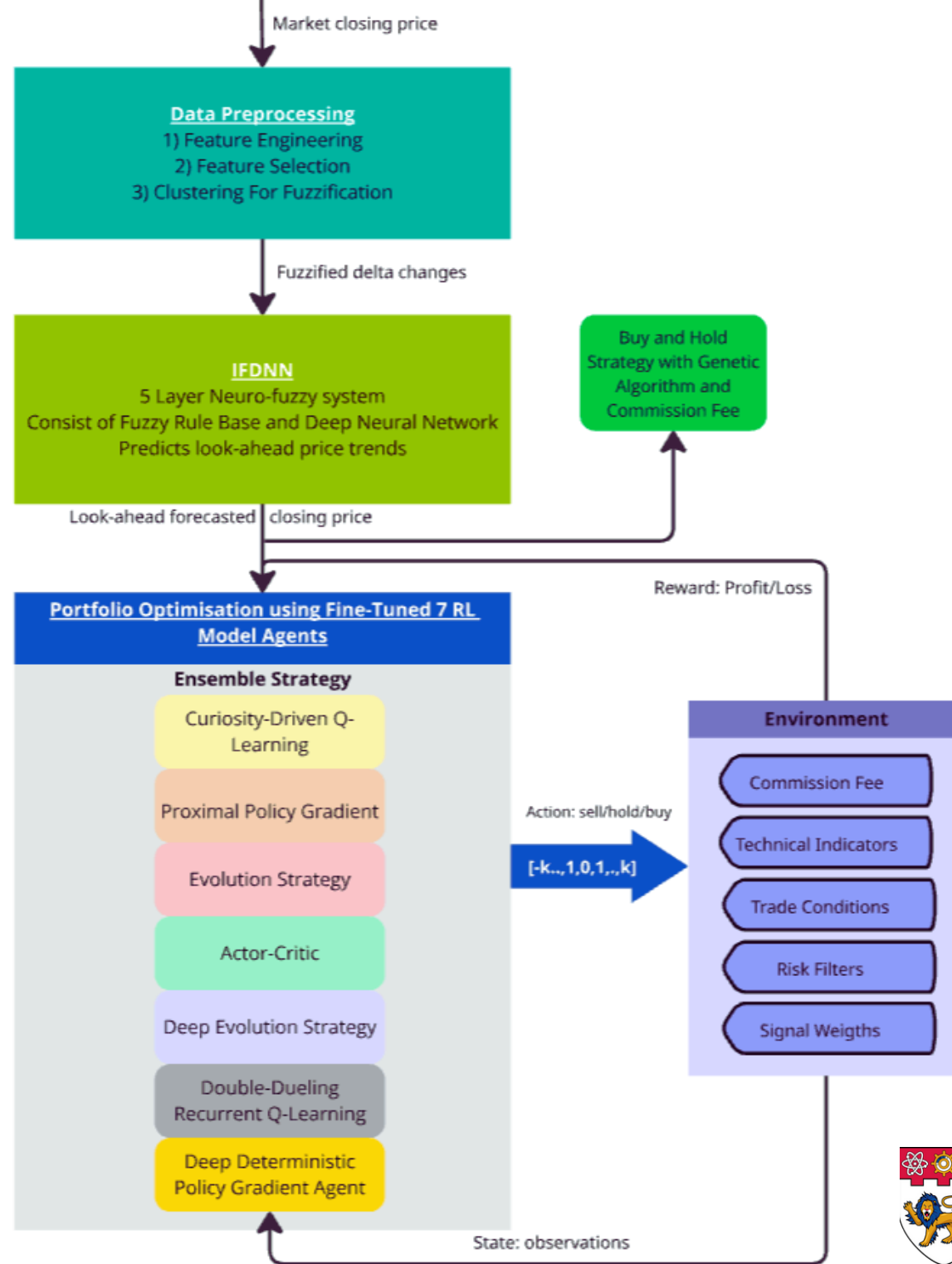
Abstract

- **Portfolio Management**

The multi-agent reinforcement learning framework was deployed to trade six SGX stocks over 2,500 days, utilizing seven optimized agents. This approach achieved a 196.31% ROI. During a high-volatility period from 2019 to 2023, with IFDTNN forecasted result, it achieved a 71.1% ROI. By dynamically selecting the best-performing agent for each stock, the framework demonstrated that no single model dominates universally.



Overview



Portfolio Selection Strategy

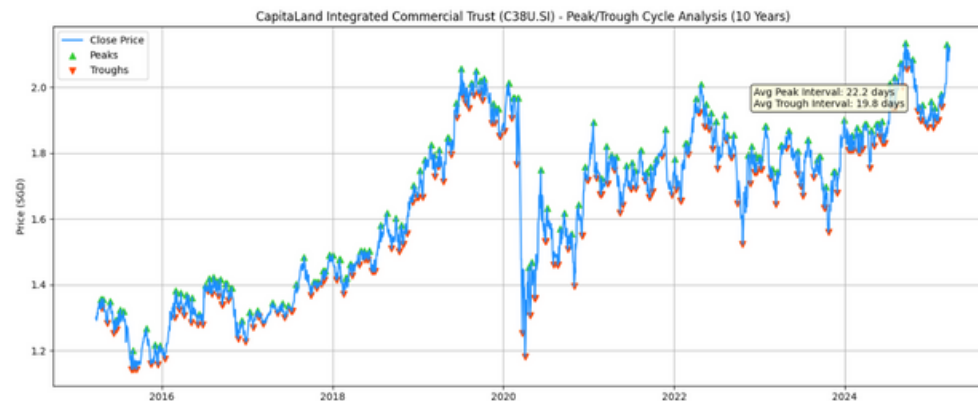
- Focused on SGX-listed stocks, SGD base currency to avoid FX risks.
- Diversified portfolio blending cyclical and defensive stocks, balancing market momentum and stability.
- Selection guided by:

Market Capitalization (large, mid, small-cap mix)

P/E Ratio (balancing value and growth picks)

Beta for lower volatility and capital preservation

Earnings Growth and Return on Equity (ROE) for operational performance



Dynamic Price Trend of C38U.SI



Data Preprocessing

Data Source:

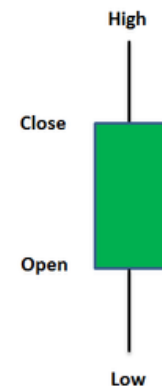
27-year historical data for 50 stocks from yfinance package

Data Includes:

Open, High, Low, Close prices (OHLC)
Trading Volume

Data Cleaning:

Handling missing values, outliers, and erroneous entries



Feature Engineering

Log Returns:

Captures percentage change in price

Price Ranges:

High-Low range each day

Momentum:

Rate of Change (ROC) of price

Volatility:

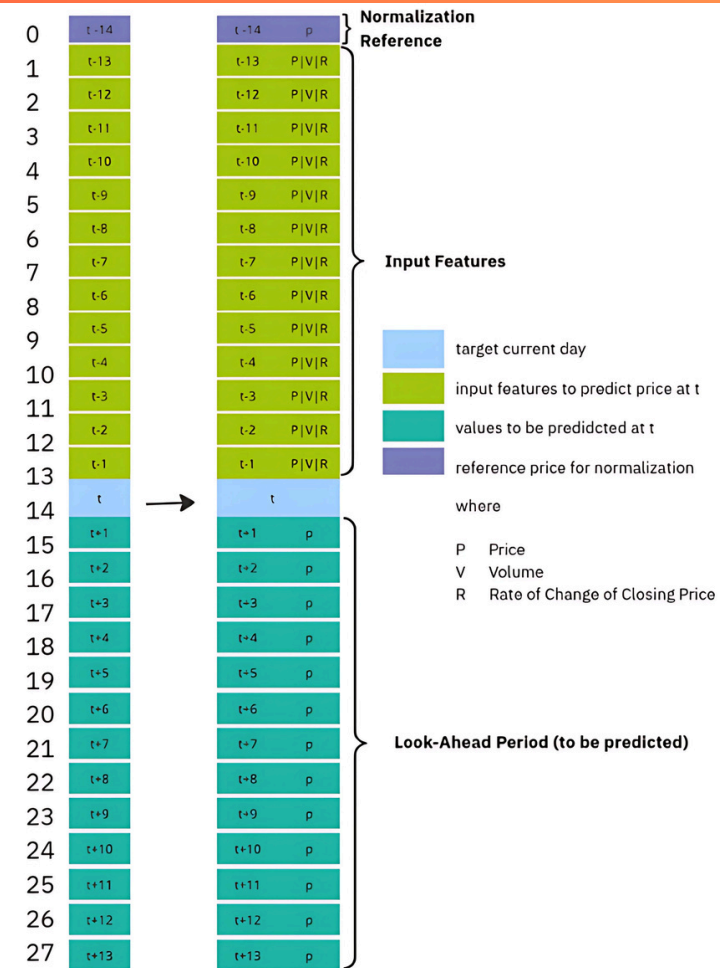
Measures daily price fluctuations

Volume:

Trading activity

Total Features:

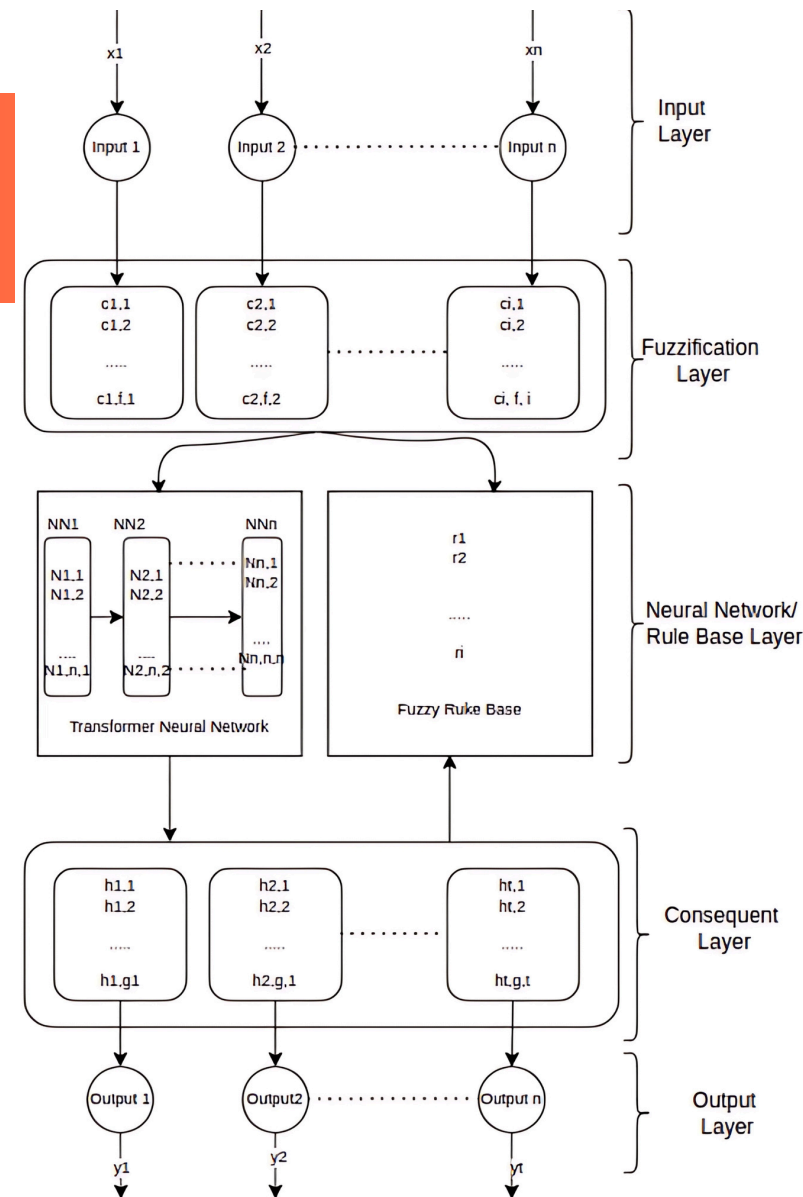
79 features per stock, including daily indicators



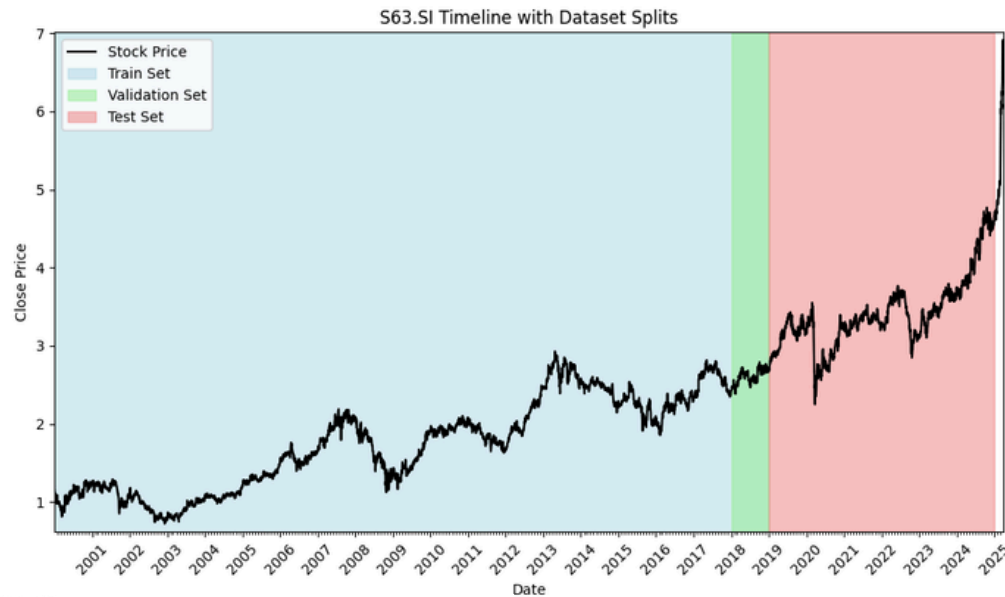
Forecasting model

Model Structure:

- **Parallel Architecture:** Transformer model works alongside the fuzzy rule base to process input data.
- **Shared Vocabulary:** Both the transformer and fuzzy systems operate using a shared vocabulary, where fuzzy rules enhance interpretability, while the transformer extracts temporal features.



Hyperparameter Optimisation

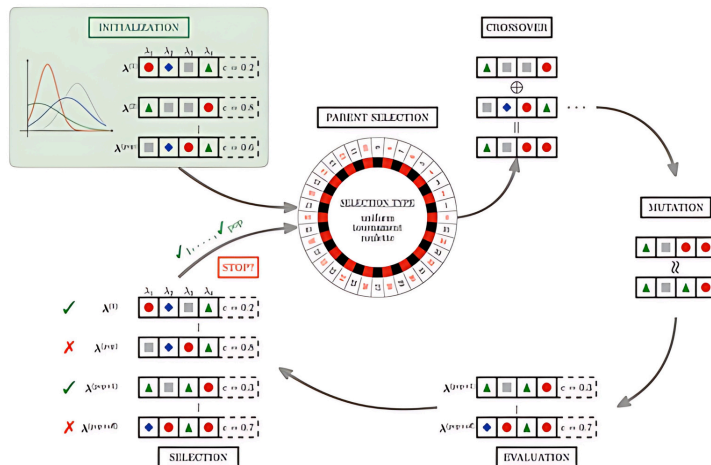


Hyperparameter	1
Hidden Layer Activation function	gelu
Encoder Dimension	256
Hidden_layers	2
Batch Size	64
Loss function	HuberLoss
Validation set RMSE	0.0436
Test R^2	0.909
Test set RMSE	0.106

Best Hyperparameter Set

The Genetic Algorithm optimization process successfully tuned the 10 hyperparameters to improve model accuracy and performance for stock price forecasting.

- Number of Attention Heads
- Number of Transformer Blocks
- Dropout Rate
- Number of MLP Layers
- Activation Function (e.g., ReLU, GELU)
- Encoder Dimension
- Learning Rate
- Batch Size
- Loss Function (e.g., HuberLoss, MSE)
- Weight Decay



Model Performance

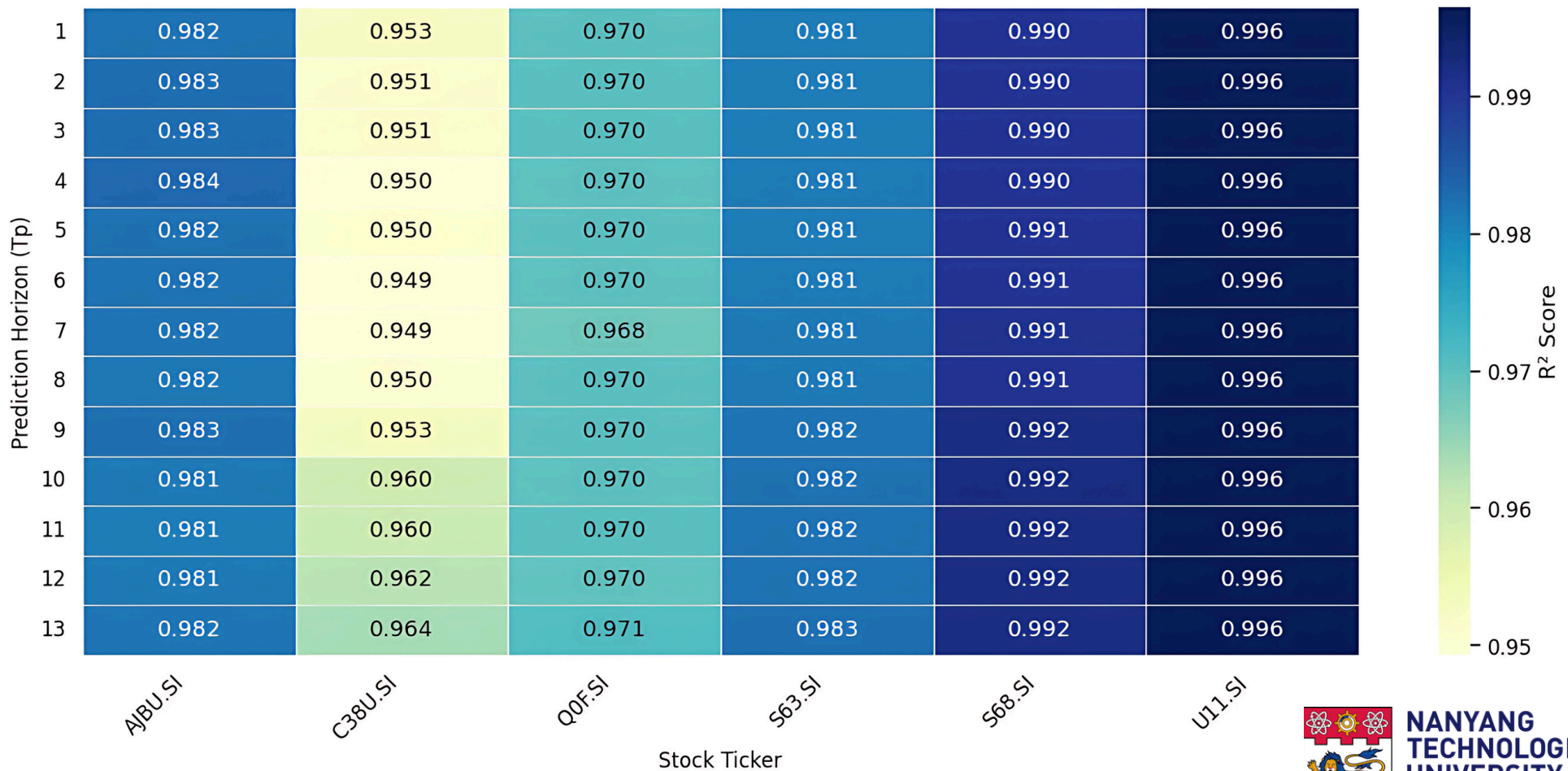
C38U.SI	(IFDTNN)		Transformer		LSTM		Bi LSTM	
Model	R2	RMSE	R2	RMSE	R2	RMSE	R2	RMSE
t+1	0.9526	0.0310	0.9553	0.1532	0.9295	0.0379	0.8343	0.0817
t+2	0.9510	0.0315	0.9560	0.2517	0.8374	0.0575	0.8204	0.1188
t+3	0.9512	0.0314	0.9546	0.1620	0.7990	0.0640	0.6607	0.0946
t+4	0.9495	0.0319	0.9344	0.1256	0.8221	0.0752	0.7628	0.0695
t+5	0.9504	0.0317	0.9322	0.1391	0.8651	0.0523	0.7913	0.0793
t+6	0.9493	0.0320	0.9341	0.1324	0.8606	0.6312	0.9038	0.0442
t+7	0.9493	0.0317	0.9333	0.1267	0.8104	0.54571	0.8632	0.0528
t+8	0.9504	0.0309	0.9378	0.1324	0.8537	0.3541	0.7126	0.0765
t+9	0.9529	0.0284	0.9421	0.1388	0.9384	0.6057	0.8649	0.0524
t+10	0.9601	0.0283	0.9564	0.1370	0.8197	0.3666	0.7043	0.0776
t+11	0.9605	0.0282	0.9451	0.1391	0.9339	0.4247	0.9369	0.0811
t+12	0.9623	0.0276	0.9358	0.1461	0.9114	0.0577	0.8284	0.0881
t+13	0.9638	0.270	0.9541	0.1463	0.8361	0,6212	0.7945	0.1016

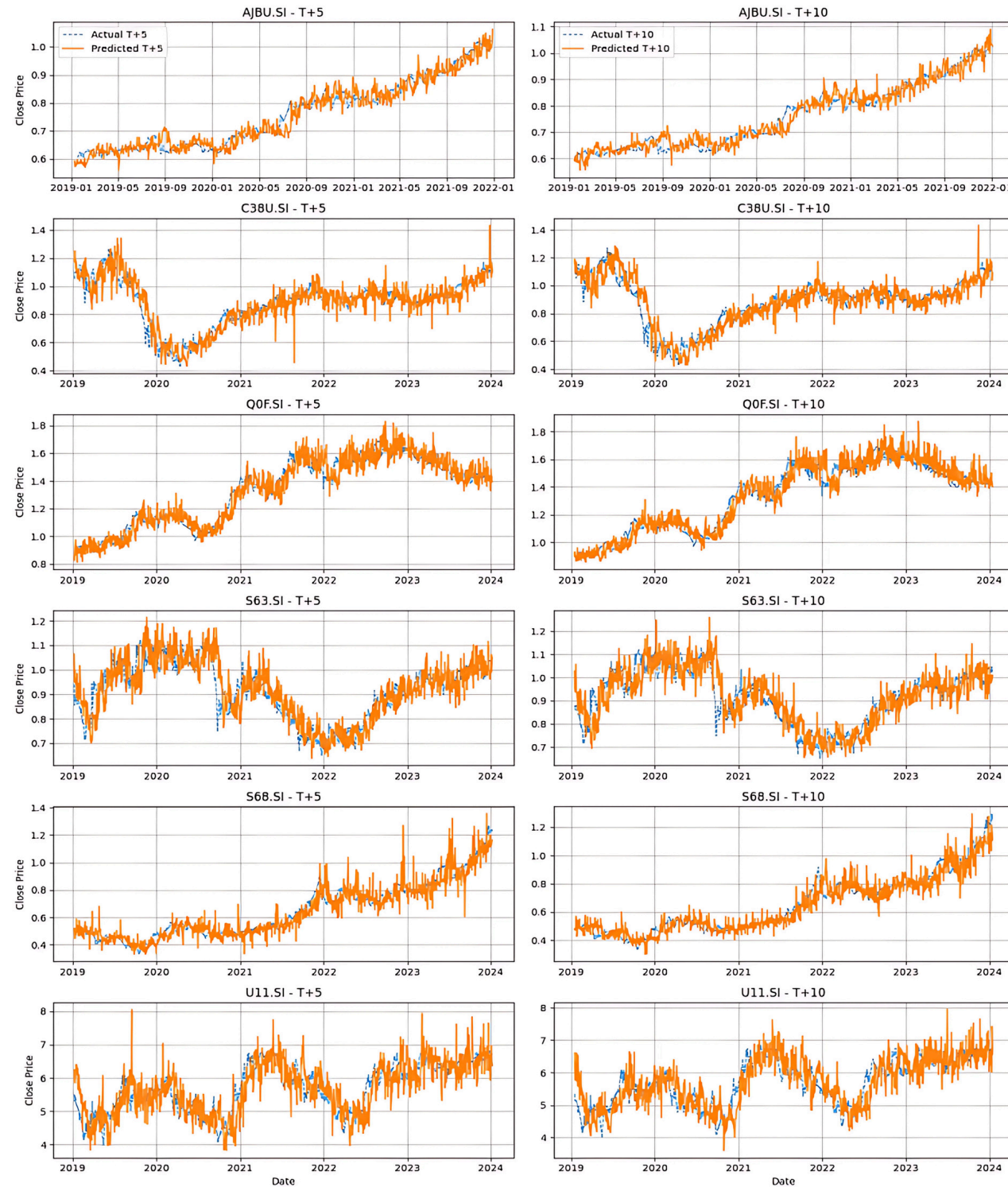
IFDTNN Benchmark on C38U.SI Crisp Feature Training Performance Comparison



Model Performance

R² Score by Ticker and Prediction Horizon (Tp1-Tp13)





Forecasting Prices



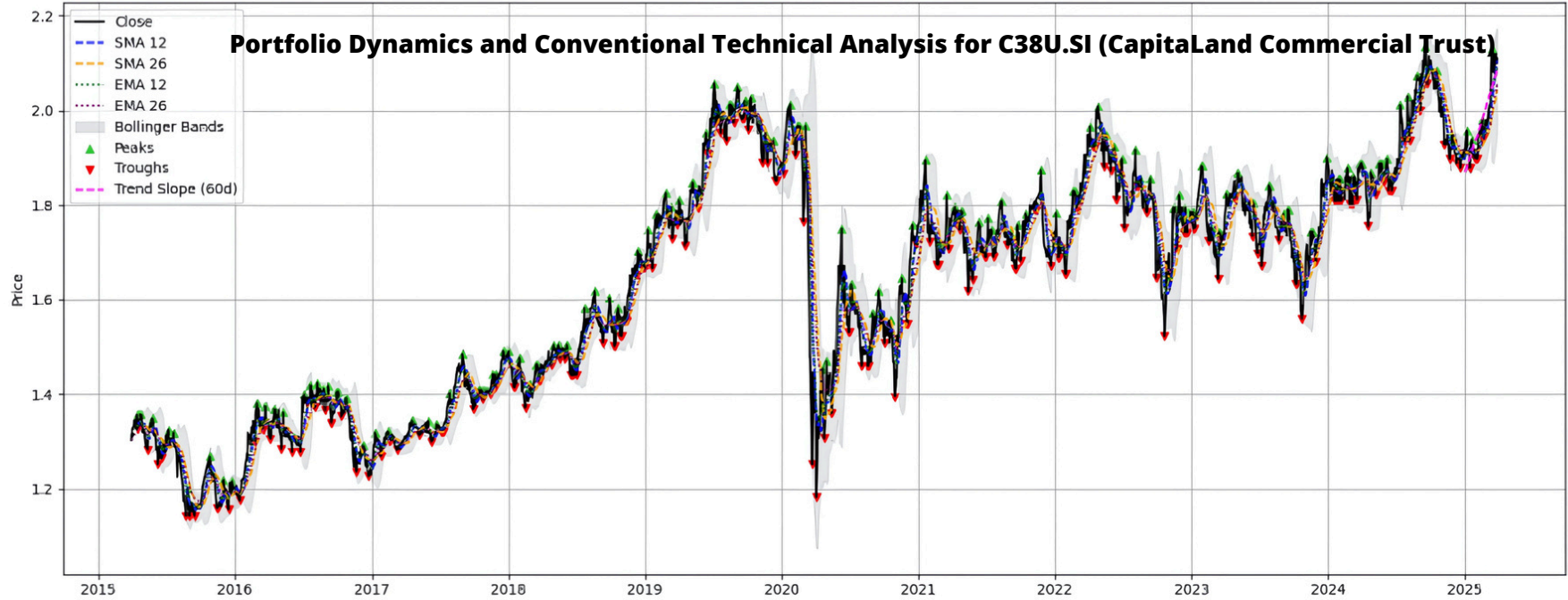
Portfolio Management

Key Trend Indictaors

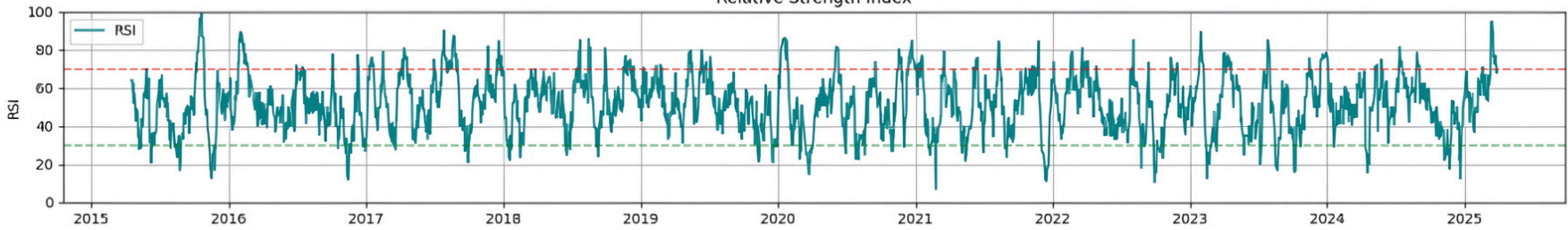
- **Simple Moving Average (SMA):** Smooths price data over a set period to reveal the market's overall trend.
- **Exponential Moving Average (EMA):** A weighted moving average that reacts faster to recent price changes.
- **Bollinger Bands:** Measures market volatility by placing bands around a moving average to indicate overbought or oversold conditions.
- **Moving Average Convergence Divergence (MACD):** Tracks the relationship between two EMAs to detect trend direction and momentum shifts.
- **Relative Strength Index (RSI):** A momentum oscillator that identifies overbought and oversold levels to gauge market strength.



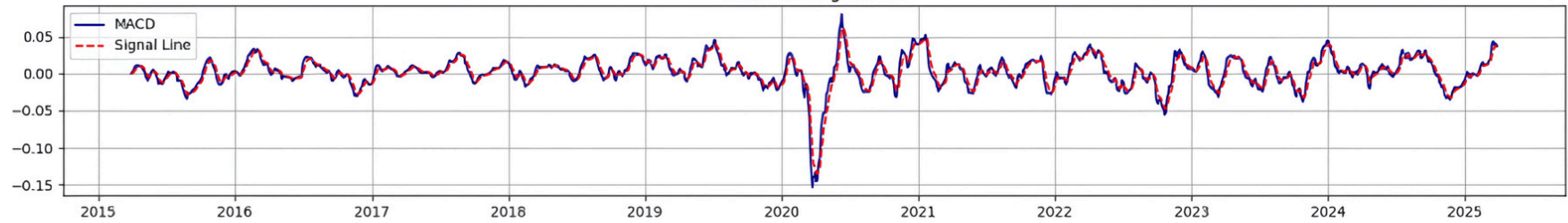
Portfolio Dynamics and Conventional Technical Analysis for C38U.SI (CapitaLand Commercial Trust)



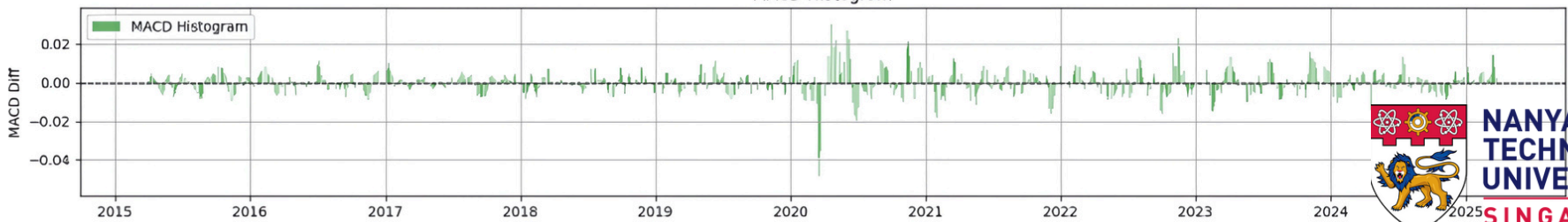
Relative Strength Index



MACD and Signal Line



MACD Histogram



Benchmark Strategies

Buy & Hold:

Baseline returns without market timing.

Buy & Hold (with commissions):

Accounts for real-world transaction costs.

Buy & Hold (with commissions) with Genetic Algorithm (GA):

Optimized buy/sell thresholds, better risk-adjusted returns.



Experiment Results

Comparative Performance Summary (Initial Investment: \$10,000 each, 2500 Days)

Ticker	Buy and Hold ROI(%) without commission	Buy and Hold ROI(%) with commission	GA incorporated Buy and Hold ROI (%) with commission
AJBU.SI	250.86	187.06	196.96
C38U.SI	62.63	33.07	88.45
Q0F.SI	29.98	6.36	65.17
S63.SI	184.61	132.85	283.24
S68.SI	139.31	95.82	141.76
U11.SI	146.91	102.16	301.30



Reinforcement Learning Agents

A2C : Actor Critic Agent

DES: Deep Evolutionary Strategy Agent

RCQL : Recurrent Curiosity Q-Learning Agent

PPO: Proximal Policy Gradient

DDRQL: Double Duel Recurrent Q-Learning Agent

DDPG: Deep Deterministic Policy Gradient Agent

ES: Evolution Strategy



Trading Strategy Parameters

- **Initial Capital:** \$10,000
- **Window Size:** 30 days
- **Commission Fee:** 0.125% per trade
- **Indicators Used:** MACD, RSI, RMA
- **Trade Conditions:**
 - Buy if weighted signal > 0.004
 - Sell if weighted signal < -0.004
- **Risk Filters:**
 - RSI: 30 (Oversold) / 70 (Overbought)
 - Volatility Buffer: 0.8%
 - Min. Profit per Trade: 0.2%
- **Signal Weights:**
 - MACD: 50%, RSI: 30%, Trend: 20%



Cross-Agent Performance

- No single agent dominates across all conditions.
- Evolutionary Strategy (ES) excels in long-term trending stocks.
- A2C and DES strong in stable, dividend-driven stocks.
- RCQL shines in volatile and momentum-driven counters.
- Multi-Agent Strategy maximized returns and reduced risk via diversification.

Cross-Agent Performance

Portfolio returns of optimized, parameter-tuned agents (initial investment of \$10,000 each) for 2500 days with commission fee

Model	A2C	DES	RCQL	PPO	DDRQL	ES	DDPG
Ticker	ROI %	ROI %	ROI %	ROI %	ROI%	ROI%	ROI %
AJBU.SI	194.75	19.36	329.69	7.13	247.07	329.70	0
C38U.SI	111.50	92.35	91.11	40.41	58.32	91.11	51.25
Q0F.SI	58.66	45.38	72.79	7.54	52.957	72.78	37.93
S63.SI	277.55	84.14	233.03	50.24	202.73	233.03	216.19
S68.SI	99.69	98.94	67.13	42.84	3.28	67.13	171.12
U11.SI	199.06	215.19	165.58	38.32	0	165.57	148.73



Overall Portfolio Performance

Ticker	Best Agent	ROI%	Final Value
AJBU	ES	329.70	42970.19
C38U.SI	A2C	111.50	21150.71
Q0F.SI	RCQL	72.79	17278.76
S63.SI	A2C	277.55	37755.69
S68.SI	DDPG	171.12	27111.89
U11.SI	DES	215.19	31518.09
Combined Portfolio Final Value : \$1777785.34 Overall ROI: 196.31			

Unpredicted Multi-agent performance of overall portfolio returns based on ROI



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IFDTNN Forecasted Result Performance During High Volatility Period (2019-2023)

Portfolio returns of optimized, parameter-tuned agents (initial investment of \$10,000 each)			
Ticker	Best Agent	ROI% (Predicted)	ROI% (Actual)
AJBU.SI	RCQL	131.41	108.85
C38U.SI	A2C	49.86	41.07
Q0F.SI	DES	34.93	29.34
S63.SI	A2C	84.31	85.76
S68.SI	DES	61.39	61.39
U11.SI	A2C	64.69	65.52
Predicted Overall ROI: 71.10% Actual Overall ROI: 65.32%			

Comparison of Performance of returns based on ROI



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Conclusion

- **Interpretable Fuzzy Deep Transformer Neural Network (IFDTNN)**
 - Transformed complex technical indicators into human-readable fuzzy rules, improving model transparency.
 - Feature selection enhanced decision-making by simplifying complex data.
- **Multi-Agent Reinforcement Learning for Portfolio Optimization using IFDTNN**
forecasted results
 - Utilized specialized RL agents for each stock, ensuring optimal strategy per asset.
 - Outperformed traditional Buy & Hold and single-agent models by integrating predictive technical indicators (RSI, RMA, EMA, MACD, Bollinger Bands), ensuring superior performance during volatile market conditions.

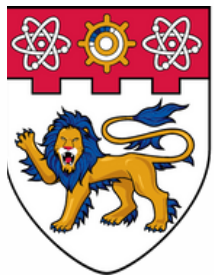


Future Works

- - Implement adaptive rebalancing strategies guided by:
 - Forecasted price movements and risks from IFDTNN
 - Dynamic threshold tuning based on market conditions
 - Implement a more robust ensemble strategy model to pick the best agents for each ticker.



**INTERPRETABLE FUZZY DEEP
TRANSFORMER NEURAL NETWORK
AND MULTI-AGENT REINFORCEMENT
LEARNING PORTFOLIO REBALANCING**



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Thank You

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